

Project Name: ConnectHub – Student Networking

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Testing and Validation

Testing Strategy

A robust testing strategy was implemented to ensure that all features in the ConnectHub platform functioned correctly and reliably. The testing phase aimed to identify defects early, verify that components integrate smoothly, and validate the end-user experience. The strategy followed a bottom-up approach, beginning with individual modules and moving toward full-system validation.

- **Unit Testing:** Focused on individual backend services such as post creation, message handling, and user authentication.
- **Integration Testing:** Checked the interaction between the frontend and backend APIs, such as how user posts are fetched and displayed.
- **System Testing:** Ensured that the entire application—from registration to logout—worked as an integrated solution.
- **User Acceptance Testing (UAT):** Involved sample users (students and professionals) who evaluated the system’s ease of use, navigation, and functionality.
- **Regression Testing:** Conducted after changes or bug fixes to ensure existing features remained unaffected.

6.2 Test Cases Overview

Test cases were prepared based on the functional requirements of the system. Each test case outlined the inputs, expected outputs, and actual results.

- **Login/Signup Test Case:** Tested for valid credentials, invalid email format, duplicate emails, and password mismatches.
- **Profile Update Test Case:** Verified whether user profile fields such as bio, skills, and role were editable and saved correctly.

- **Post Creation/Editing Test Case:** Ensured that users could create, update, and delete posts. Also checked for validation messages when fields were left blank.
- **Follow Topic/User Test Case:** Verified the ability to follow and unfollow users or topics, with real-time updates to the feed.
- **Admin Panel Test Case:** Included testing of functionalities like deleting posts, viewing reports, and deactivating users

Test Case ID	Test Scenario	Input	Expected Result	Actual Result	Status
TC-01	User Registration/Login	Valid/invalid email & password	Successful login or error message on invalid credentials	Works as expected	Pass
TC-02	Profile Update	New bio, skills, and role	Profile updated and changes reflected	Changes saved correctly	Pass
TC-03	Post Creation	Text input with/without topic	Post appears in feed or error if required field is empty	Works as expected	Pass
TC-04	Post Editing/Deletion	Edited post content or delete action	Post updated or removed from feed	Functions correctly	Pass
TC-05	Follow Topic or User	Click on "Follow" button	Topic/user added to following list	List updated instantly	Pass

Test Case ID	Test Scenario	Input	Expected Result	Actual Result	Status
TC-06	Messaging Between Users	Message text, recipient ID	Message sent and displayed in chat inbox	Delivered successfully	Pass
TC-07	Admin Delete Post	Select and delete reported post	Post is removed from database and UI	Action completed	Pass
TC-08	Form Validation	Blank fields, invalid email	Error messages displayed, form not submitted	Validation works	Pass
TC-09	Cross-Browser Compatibility	Load site on Chrome, Firefox, Safari	Uniform layout and full functionality	UI consistent across all	Pass
TC-10	Mobile Responsiveness	Load site on phone/tablet	Responsive layout with all core features	Fully responsive	Pass

Validation Techniques

To ensure the final system aligns with user needs and functions effectively in real-world conditions, multiple validation techniques were applied:

- **Requirement Traceability Matrix (RTM):** Used to confirm that all specified requirements were implemented and tested.
- **Edge Case Validation:** Tested system behavior under rare or unexpected conditions, such as special characters in usernames or long message content.

- **Form Validation:** Ensured mandatory fields were filled and incorrect formats were caught in real-time (e.g., email validation).
- **Device and Browser Testing:** Confirmed that the platform performed consistently across Chrome, Firefox, and mobile browsers.
- **Feedback Loops:** User feedback collected during testing cycles was used to improve UI, messaging clarity, and performance.

6.4 Result Summary

After comprehensive testing, the results indicate that ConnectHub meets both its functional and non-functional requirements effectively.

- The system allows for secure login, post sharing, topic following, and peer communication.
- No critical bugs were found during final testing. Minor issues (e.g., input validation errors, UI misalignments) were resolved quickly.
- Performance under standard internet connectivity was stable, and user interactions were responsive.
- Test users expressed satisfaction with the clean interface, ease of use, and the research-focused design.

These results validate that ConnectHub is ready for wider deployment and potential scaling to more institutions and professionals.